analyzed the chemical composition of DNA isolated from different species and found that irrespective of the source, the molar concentration of Adenine always equals Thymine and the molar concentration of Guanine always equals Cytosine. A=T and G=C. Therefore, A + T = G + C and the ratio of A+T /G+C = 1.0,which means that the total number of purine bases = the total number of pyrimidine bases.

## 7.6 Denaturation of DNA

At high temperatures (95$^0$C), the double helical structure of DNA melts due to disruption of base pairing that results in two single strands. This is called as Denaturation of DNA. The temperature at which it does so is called as Melting temperature (Tm). AT rich regions melt faster than GC rich regions. Therefore, Tm is dependent upon the composition of DNA. During denaturation, the absorption of DNA at 260nm increases. This property of DNA is called as hyperchromicity. If the temperature is brought down, the single strands rejoin to form double stranded regions. This is called as annealing of DNA (Fig. 7.8). This property of DNA is exploited in Polymerase Chain Reaction.

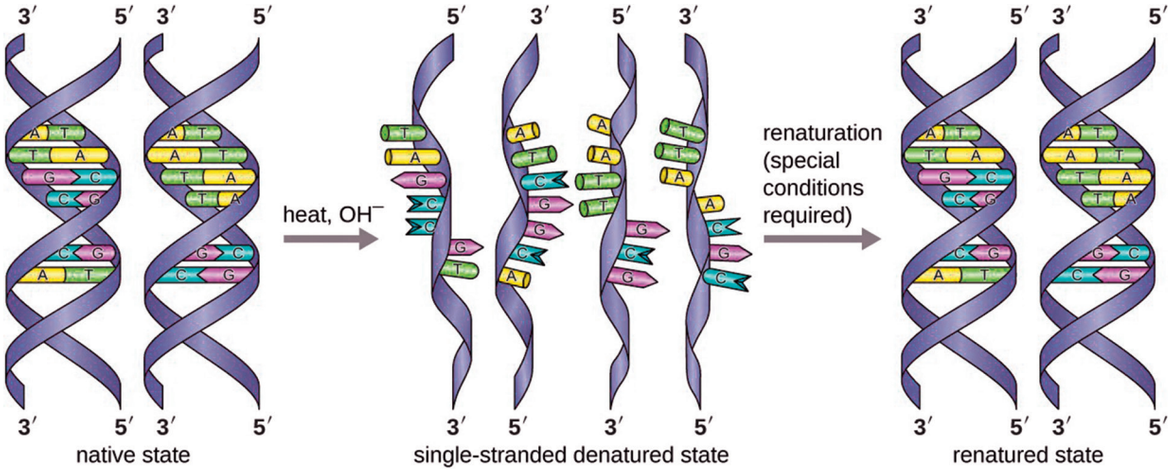



*Figure. 7.8 Denaturation and renaturation of DNA*

## 7.7 Griffith's experiment to identify the genetic material

Frederick Griffith conducted experiments in 1928 with *Streptococcus pneumoniae*, which was the milestone for discovery of DNA as the genetic material. There are two different strains of *Streptococcus pneumoniae*, one with smooth shiny colonies called the S strain, while the other with rough colonies (R). The difference is due to the presence of mucous coat in S strain bacteria, whereas the R strain bacteria lacked them. Because of the presence of the mucous coat, S strain is lethal, while R strain is not lethal.

Griffith injected both S and R strains to two different groups of mice separately. The group which was infected with the S strain developed pneumonia and died while that infected with the R strain stayed alive.



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